Noise-whitened VarPro fitting: combining smooth and discrete basis functions without mass matrices in the fit

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Abstract

Per-row VarPro fitting of the LG-PSF Hessian model combines two kinds of basis function: theta-dependent smooth (Laguerre–Gaussian) features, and a fixed theta-independent "extra" basis (a discrete spike, and generalizations of it) supplied directly as sampled values. The two live in different spaces with different natural inner products (one in the discretized function space, one in its dual), and combining them naively – e.g. orthogonalizing them against each other under a plain Euclidean inner product – either gets the wrong answer or requires the fitting code to know about the mesh mass matrices explicitly. This note derives a *noise-whitening* change of variables under which the entire per-row fit becomes an ordinary (mass-free) least-squares problem: every basis function, every derivative, and the data itself are rescaled once, at the interface, so that plain Euclidean operations throughout the fit are automatically correct. The two mass matrices (M_1 for the row/target space, M_2 for the column/source space) appear only in that one whitening layer; the VarPro fitting machinery itself never sees them. The whitening transform is theta-independent and symmetric, so it composes with the existing forward/reverse-mode derivative machinery (`lg_functions.hpp`, `ellipsoid_transform.hpp`) with no new derivative math required.

1 Setup

Recall the discretization picture: X is the discrete source/column function space with inner product $\langle u, v\rangle_X = u^\top M_2 v$; Y is the discrete target/row space with $\langle u, v\rangle_Y = u^\top M_1 v$; X', Y' are their duals, with $M_2 : X \to X'$ and $M_1 : Y \to Y'$ the Riesz maps induced by those inner products; the kernel $\Phi : X' \to Y$ and the assembled operator $H = M_1 \Phi M_2 : X \to Y'$. Both M_1, M_2 are diagonal (lumped mass).

Each Laguerre–Gaussian smooth feature $\phi_i(\cdot;\theta)$, sampled at mesh nodes, is an element of X – and, because the continuum family $\{\psi_i\}$ is exactly $L^2(\mathbb{R}^N)$-orthonormal (verified numerically in `test_lg_functions.cpp` by tensor-product Gauss–Hermite quadrature, which pins the generated harmonic table, the Laguerre recurrence and the normalization at once), the sampled ϕ_i are nearly M_2-orthonormal: $\phi_i^\top M_2 \phi_{i'} \approx \delta_{ii'}$, up to mesh resolution. A user-supplied "extra" basis function e_d (a one-hot vector for the diagonal spike; more general indicator combinations for e.g. a ring-neighbor correction) is instead a sampled *functional* – the coordinate representation of a discrete measure (a sum of point masses) via the FEM nodal interpolation property $N_k(x_i) = \delta_{ik}$ – and so is native to X', not X.

2 The row model

Fix a row ρ with mass $m_\rho := (M_1)_{\rho\rho}$, and a local neighbor/support batch of K column points x_j with per-point masses $m_j := (M_2)_{jj}$. The row model is

$$H[\rho, j] \;\approx\; \underbrace{m_\rho m_j \sum_i c_i\, \phi_i(x_j;\theta)}_{\text{smooth}} \;+\; \underbrace{m_\rho \sum_d s_d\, e_d(j)}_{\text{extra}}, \tag{1}$$

with c_i, s_d the (linear) coefficients VarPro solves for at each trial θ. The column mass m_j on the smooth term is a quadrature weight – ϕ_i approximates a continuum kernel evaluated at a mesh quadrature point, and m_j discretizes the integral. The extra term carries no quadrature weight because it is not a continuum-kernel evaluation at all: it is a direct, discrete correction to specific matrix entries. This asymmetry is what reproduces the $m_\rho s_\rho \delta_{\rho j}$ spike convention exactly.

Probing: draw i.i.d. standard normal z_ℓ ($\ell = 1, \ldots, k$, restricted to the batch – a restriction of an i.i.d. vector is still i.i.d.), observe $y_\ell(\rho) = (Hz_\ell)(\rho) = \sum_j H[\rho, j]\, z_\ell(j)$. Substituting (1),

$$y_\ell(\rho) \;=\; m_\rho \sum_i c_i\, g_\ell^{(i)} \;+\; m_\rho \sum_d s_d\, h_\ell^{(d)}, \qquad g_\ell^{(i)} := (M_2\phi_i)\cdot z_\ell, \quad h_\ell^{(d)} := e_d \cdot z_\ell. \tag{2}$$

The problem with combining these naively. ϕ_i (in X) and e_d (in X') are different kinds of object; comparing or orthogonalizing them requires first mapping one into the other's space via the Riesz map, and then using *that* space's natural inner product. Using the raw Euclidean inner product throughout happens to reproduce the correct answer for a single one-hot e_d – every diagonal-respecting inner product agrees on a coordinate-axis-aligned vector – but is not correct in general (e.g. a ring-neighbor indicator combining several points), and more importantly requires the fitting code to reach into M_2 (and its inverse) explicitly to do the orthogonalization correctly. That is exactly the kind of mass-matrix leakage this note is trying to eliminate.

3 Noise whitening

Define, per row (all quantities below implicitly depend on ρ through m_ρ and the batch):

$$\widehat{z}_\ell := M_2^{1/2} z_\ell, \tag{3}$$

$$\widehat{\phi}_i(\theta) := \sqrt{m_\rho}\; M_2^{1/2} \phi_i(\theta), \tag{4}$$

$$\widehat{e}_d := \sqrt{m_\rho}\; M_2^{-1/2} e_d, \tag{5}$$

$$\widehat{y}_\ell := y_\ell(\rho)/\sqrt{m_\rho}. \tag{6}$$

$M_2^{1/2}$ is diagonal, hence symmetric, so $(M_2\phi_i)\cdot z_\ell = (M_2^{1/2}\phi_i)\cdot(M_2^{1/2} z_\ell)$, giving $\widehat{\phi}_i \cdot \widehat{z}_\ell = \sqrt{m_\rho}\, g_\ell^{(i)}$; likewise $\widehat{e}_d \cdot \widehat{z}_\ell = \sqrt{m_\rho}\,(M_2^{-1/2} e_d)\cdot(M_2^{1/2} z_\ell) = \sqrt{m_\rho}\, e_d \cdot z_\ell = \sqrt{m_\rho}\, h_\ell^{(d)}$. Dividing (2) by $\sqrt{m_\rho}$:

$$\widehat{y}_\ell \;\approx\; \sum_i c_i \big(\widehat{\phi}_i(\theta)\cdot \widehat{z}_\ell\big) \;+\; \sum_d s_d \big(\widehat{e}_d \cdot \widehat{z}_\ell\big). \tag{7}$$

This is an ordinary (unweighted) linear regression in ℓ: every mass matrix has been absorbed into the four definitions (3)–(6), and (7) itself is mass-free.

Note on $\widehat{e}_d$: M_1 contributes the same $\sqrt{m_\rho}$ factor to *both* terms of (1) (there is one row mass, shared), so $\widehat{e}_d$ needs the same $\sqrt{m_\rho}$ prefactor as $\widehat{\phi}_i$ even though it does not involve M_2 (which enters with the opposite sign of exponent, $-1/2$ vs. $+1/2$, reflecting that e_d lives in X' where ϕ_i lives in X). This factor is easy to drop, and dropping it is invisible to every check that does not touch M_1 or the extra basis – finite differences and adjoint consistency on the feature layer both pass without it. `test_whitening.cpp` guards it directly, by constructing an explicit H from (1) and checking that the whitened quantities reproduce $y_\ell(\rho)$ to machine precision.

4 Consequences

Orthonormality is inherited directly. $\widehat{\phi}_i \cdot \widehat{\phi}_{i'} = m_\rho\,(M_2^{1/2}\phi_i)\cdot(M_2^{1/2}\phi_{i'}) = m_\rho\,\phi_i^\top M_2 \phi_{i'} \approx m_\rho\,\delta_{ii'}$: the whitened smooth basis vectors are, in the ordinary Euclidean sense, exactly as orthonormal (up to the constant m_ρ) as the M_2-orthonormality already established for the sampled LG modes. No separate weighted inner product is needed to see this – it falls out of using the plain dot product on already-whitened vectors.

Plain-Euclidean orthogonalization is now correct by construction. Projecting a smooth feature against an extra basis correctly requires the M_2^{-1}-weighted inner product on X': $\widehat{\phi}_p^\flat \mapsto \widehat{\phi}_p^\flat - e_i\,(e_i^\top M_2^{-1} e_i)^{-1}(e_i^\top M_2^{-1}\widehat{\phi}_p^\flat)$, $\widehat{\phi}_p^\flat := M_2\phi_p$. Checking this against the whitened quantities here: $\widehat{\phi}_p\cdot\widehat{e}_i = (M_2^{1/2}\phi_p)\cdot(M_2^{-1/2}e_i) = \phi_p\cdot e_i$ and $\widehat{e}_i\cdot\widehat{e}_i = e_i^\top M_2^{-1} e_i$ (the $\sqrt{m_\rho}$ factors cancel in the ratio) – identical projection coefficients. So plain-Euclidean orthogonalization of $\widehat{\phi}_p$ against $\widehat{e}_i$ *is* the earlier, carefully-derived M_2^{-1}-weighted projection; whitening does not introduce a different answer, it removes the need to ever justify or implement a non-Euclidean inner product near the fitting code.

Derivatives compose for free. $\sqrt{m_\rho}\,M_2^{1/2}$ and $\sqrt{m_\rho}\,M_2^{-1/2}$ are fixed, θ-independent, symmetric linear operators (diagonal matrices). Consequently:

- the JVP of $\widehat{\phi}_i$ is that operator applied to the JVP of the raw $\phi_i(\theta)$ – i.e. apply it to whatever `grad_eval_lg_nd` · `jvp_T` (chain rule through the pullback) already produces;
- because the operator is symmetric, the VJP is the *same* operator applied to the incoming cotangent *before* it is fed into the existing VJP chain.

This is the same "compose with a fixed linear map" pattern that `ellipsoid_transform.hpp` already uses to chain its two stages. No new derivative formulas are needed anywhere.

5 Resulting architecture

- **Mass-free composed feature layer** (`lg_ellipsoid_feature.hpp`): $\phi_i(x;\theta) := \psi_i(T(\theta,x))$ and its JVP/VJP, built purely by composing the two existing modules via the chain rule. Knows nothing about mass matrices.
- **Whitening interface layer** (`whitening.hpp`): the only place M_1, M_2 appear. Provides (a) `whiten_probes`, `whiten_data`, `whiten_extra` for the user-supplied raw probe/data/extra-basis arrays, and (b) whitened wrappers around the feature layer's eval/JVP/VJP, applying $\sqrt{m_\rho}\,M_2^{\pm 1/2}$ once.

- **Generic VarPro core** (`varpro.hpp`): takes whitened probes, whitened data, and a basis functor $\theta \mapsto$ an evaluation offering values and a VJP. Performs the per-row fit in ordinary Euclidean terms throughout and never references a mass matrix. Concretely it projects the extra block out once by Frisch–Waugh–Lovell rather than at every trial point, solves the inner least-squares problem, forms the reduced residual, and hands its θ-Jacobian to a trust-region Levenberg–Marquardt loop (`detail/levenberg_marquardt.hpp`).

The prediction that this layering would hold has since been tested by building on top of it. Two things are worth recording because they confirm it.

First, the fitting core is genuinely generic: `varpro.hpp` is templated on the basis functor and contains nothing specific to Laguerre–Gaussians, ellipsoids or point-spread functions. It is usable for any separable least-squares problem.

Second, the whitening layer stayed exactly one file. Every mass matrix in the library still appears only in `whitening.hpp`, and everything above it — the candidate search, the mode ladder, the cross-validation scoring, the whole-operator layer — works in whitened coordinates without ever needing M_1 or M_2. The one place a mass reaches higher is a scalar: the operator layer passes each row's own m_ρ down as `target_mass`, so the returned coefficients come back in physical units. Because the whitened design matrix is column-equilibrated inside the inner solve, that scalar rescales only (c, s) — the fitted θ, the scores and the entire selection are invariant to it.

6 Operator-level quality control

The fit makes two different kinds of decision, and they deserve two different metrics. The per-row cross-validation score selects a *model for one row* (how many LG modes to use), and it already operates in whitened coordinates: its residuals are $\widehat{y}$-residuals, i.e. residuals in the $y/\sqrt{m_\rho}$ scaling of (6). Nothing in this section touches it. The second decision is made about the *operator as a whole* – is the assembled fit $B \approx H$ accurate enough to stop spending probes? – and this section derives the metric for that decision, extending the whitening story from rows up to the operator.

Why a naive global metric misleads. A natural-looking choice is to hold out a few probes and average their relative residuals, $\frac{1}{k}\sum_\ell \|Bz_\ell - y_\ell\|/\|y_\ell\|$. The trouble is the denominator: $\|Hz\|$ is dominated by the overlap of z with the leading spectral directions of H, and the operators we care about have large spectral outliers – that is the very reason they need preconditioning. Each per-probe ratio therefore has a heavy-tailed denominator, and with a handful of held-out probes the *mean of ratios* inherits the tail: its value is largely a property of the draw, not of the fit. (In the ice-sheet application we observed a factor-of-three spread across probe draws at fixed fit quality.) The fix is the same one used throughout these notes: state the quantity in the natural norms, whiten, and then let plain Euclidean arithmetic be correct by construction.

The metric. The error $D := H - B$ maps $X \to Y'$, so its natural Hilbert–Schmidt norm measures inputs in $\langle u, v\rangle_X = u^\top M_2 v$ and outputs in the dual norm $\|r\|^2_{Y'} = r^\top M_1^{-1} r$. Summing $\|Du_k\|^2_{Y'}$ over an X-orthonormal basis $\{u_k\}$ gives

$$\|D\|^2_{\mathrm{HS}(X,Y')} \;=\; \left\|M_1^{-1/2}\,(H - B)\,M_2^{-1/2}\right\|_F^2. \tag{8}$$

That is, the natural operator error is the plain Frobenius norm of the error after whitening both sides – the operator-level analogue of the per-row transforms (3)–(6), with the single row mass m_ρ replaced by the full M_1. The quality-control quantity is the relative version, $\|D\|_{\mathrm{HS}}/\|H\|_{\mathrm{HS}}$.

The estimator. Draw held-out probes i.i.d. standard normal *in the whitened variables*: $\widehat{z}_\ell \sim N(0, I)$, i.e. $z_\ell = M_2^{-1/2}\widehat{z}_\ell$ in raw coordinates, so that $\mathbb{E}\, z_\ell z_\ell^\top = M_2^{-1}$. In function-space terms this makes z_ℓ white noise with respect to the X inner product – coefficient covariance M_2^{-1} is the FEM representation of L^2-white noise – whereas a coordinate-i.i.d. draw sits one factor of M_2 away from it. Then

$$\mathbb{E}\left\|M_1^{-1/2}(H-B)z_\ell\right\|^2 \;=\; \|D\|^2_{\mathrm{HS}(X,Y')}, \qquad \mathbb{E}\left\|M_1^{-1/2}Hz_\ell\right\|^2 \;=\; \|H\|^2_{\mathrm{HS}(X,Y')}, \tag{9}$$

and the quality control is the ratio of sums (an *energy* ratio, not a mean of ratios):

$$\mathrm{qc}^2 \;=\; \frac{\sum_\ell \left\|M_1^{-1/2}(Bz_\ell - y_\ell)\right\|^2}{\sum_\ell \left\|M_1^{-1/2}y_\ell\right\|^2}, \qquad y_\ell = Hz_\ell. \tag{10}$$

Two remarks on this estimator. First, as a ratio of sums it is biased at $O(1/k)$ for the ratio of expectations, which is immaterial for a stopping test. Second, and decisively, it concentrates where the mean of ratios does not: numerator and denominator fluctuate *together* through the shared draw, so their ratio is far more stable than either sum alone. The heavy tail that poisons the per-probe ratios enters both sums the same way and largely cancels.

Held-out interpretation without the i.i.d. assumption. The i.i.d.-in-whitened-variables assumption is what upgrades (10) to an unbiased estimate of the *global* relative Hilbert–Schmidt error; it is not what makes the formula meaningful. For any held-out probe family – scaled or unscaled, random or structured (polynomials, sinusoids, Dirac combs, draws from other smoothness classes) – (10) is still a perfectly reasonable held-out relative error: the error of B's action on that family, measured in the natural norms. What changes with the probe family is only *which* operator error the number reflects – the family weights the directions being tested – not whether the number is a valid relative error on them.

Mesh independence. Whitening is what makes the number mean the same thing on different meshes. Heuristically, (8) is the quadrature approximation of the continuum Hilbert–Schmidt (relative) norm of the kernel error, so under refinement – uniform or adaptive – the metric approaches a mesh-independent limit whenever the underlying kernel is square-integrable. We do not prove this; it is the standard quadrature heuristic. By contrast, the unweighted version (Euclidean norms, coordinate-i.i.d. probes) estimates the raw-coordinate Frobenius ratio, which weights every node equally: on an adaptive mesh, refining a region silently re-weights the metric toward it. On a quasi-uniform mesh the two versions differ only by constants that cancel in the ratio, which is why the distinction is easy to miss until an adaptive mesh makes it matter.

What changes and what does not. In whitened variables the probing rule is unchanged – "draw $\widehat{z}$ i.i.d." is exactly the (3) convention – and the residual weighting $M_1^{-1/2}$ is the global form of the per-row $y/\sqrt{m_\rho}$. The per-row scores, the mode selection, and the fitted operator are untouched. Two consequences do warrant attention. First, in raw coordinates the probes

acquire a factor $M_2^{-1/2}$, and the fit consumes the same probes as the quality control: changing the probe covariance changes the fitted operator (incidentally, it makes the whitened per-row design itself i.i.d., since $\widehat{z} = M_2^{1/2} z =$ the drawn vector), so adopting (10) is a change to the fit's inputs, not only to a printed number. The per-row fits are insensitive to this change: a row's regression is valid conditioned on whatever probes were drawn – the probe distribution affects statistical efficiency, not correctness – and within one PSF window the column masses vary mildly, so the covariance change amounts to a near-constant factor per window, which the regression absorbs. It is only the *global* quantity, on a mesh whose resolution varies by orders of magnitude across the domain, that needs the scaling. Second, any threshold calibrated against a previous metric must be recalibrated against this one, once.